
Alternativa

a) (1 punt) Calcula $\int \frac{x^2+1}{x} dx$ i $\int \frac{(x+1)^2}{\sqrt{x}} dx$, separant cada quocient en una suma.

Solució. $\frac{x^2+1}{x} = x + \frac{1}{x}$, i $\int \frac{x^2+1}{x} dx = \frac{x^2}{2} + \ln|x| + k$.

$$\frac{(x+1)^2}{\sqrt{x}} = \frac{x^2+2x+1}{x^{1/2}} = x^{3/2} + 2x^{1/2} + x^{-1/2}, \text{ i}$$

$$\int \frac{(x+1)^2}{\sqrt{x}} dx = \frac{2}{5}x^{5/2} + \frac{4}{3}x^{3/2} + 2x^{1/2} + k = \frac{2}{5}x^2\sqrt{x} + \frac{4}{3}x\sqrt{x} + 2\sqrt{x} + k.$$